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TOPICAL REVIEW

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Topical Review

Design, fabrication and application of magnetically actuated micro/nanorobots: a review

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Abstract

Magnetically actuated micro/nanorobots are typical micro- and nanoscale artificial devices with favorable attributes of quick response, remote and contactless control, harmless human-machine interaction and high economic efficiency. Under external magnetic actuation strategies, they are capable of achieving elaborate manipulation and navigation in extreme biomedical environments. This review focuses on state-of-the-art progresses in design strategies, fabrication techniques and applications of magnetically actuated micro/nanorobots. Firstly, recent advances of various robot designs, including helical robots, surface walkers, ciliary robots, scaffold robots and biohybrid robots, are discussed separately. Secondly, the main progresses of common fabrication techniques are respectively introduced, and application achievements on these robots in targeted drug delivery, minimally invasive surgery and cell manipulation are also presented. Finally, a short summary is made, and the current challenges and future work for magnetically actuated micro/nanorobots are discussed.

Keywords: micro/nanorobots, magnetic actuation, contactless control, bio-robots

(Some figures may appear in colour only in the online journal)

1. Introduction

Micro/nanorobotics is a burgeoning interdisciplinary field integrating micro/nano technology, robotics and biomedical technology, which has been highly favored by researchers in recent years [1]. Compared with macroscale robots, micro/nanorobots have many unique advantages of small size, light weight and high dexterity, and they are more suitable for biomedical applications such as targeted drug delivery, *in vivo* detection, minimally invasive surgery and cell manipulation [2–4]. Micro/nanorobots perform predetermined tasks in some complex microscopic liquid environments through different propulsion mechanisms achieved by a variety of actuation strategies [5]. Actuation strategies mainly include physical actuation (e.g. magnetic field, electric field, light field, thermal field, ultrasonic wave and piezoelectric effect), chemical actuation (e.g. chemical reactions) and biological actuation (e.g. motile cells and microorganisms). Each actuation strategy is designed to match the functionalities of micro/nanorobots to their application environments.

Although great advances of micro/nanorobots based on the various actuation modes described above have been made, each of these actuation methods has some drawbacks. The electrodes of electrical actuation limit the robot's working areas and biomedical applications [6–8]. The heat generated by the light actuation maybe damage active biological

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materials and human tissues [9–11]. Similarly, thermal actuation maybe cause permanent damage to living tissues or organs [12–14]. It is difficult for micro/nanorobots based on ultrasonic actuation to work in some environments containing hard tissues and gases [15–17]. Piezoelectric actuation is usually suitable for manufacturing larger robots (e.g. sub-millimeter robots and millimeter robots), which makes the application range of the robots based on piezoelectric actuation narrow [18, 19]. Due to the uncontrollable chemical reaction process, the biological stability and motion accuracy of the chemically driven micro/nanorobot are poor. Moreover, the short chemical reaction time limits the effective working time of such micro/nanorobots [20–22]. Although biological actuation has the excellent biocompatibility, it is necessary to create a working environment suitable for the living cells or microorganisms used for fabrication of biohybrid robots to remain active for a long time [23–25].

Among the above actuation strategies, magnetic actuation is a contactless energy conversion and the low-strength magnetic field is medically proven harmless to biological bodies. The magnetic field penetrates biological bodies by a noninvasive means to manipulate matter based on the spatial non-contact properties of magnetic forces [26]. Owing to their excellent attributes of quick response, remote and contactless control, harmless human-machine interaction and high economic efficiency [5], magnetically actuated micro/nanorobots are recognized as one of the most promising micro/nanorobots for biomedical applications. In view of this, a large number of magnetically actuated micro/nanorobots have been developed in recent years to address current challenges encountered by targeted drug delivery, minimally invasive surgery and cell manipulation.

Generally, magnetically actuated micro/nanorobots usually work in non-Newtonian fluid environments (e.g. blood, lymph, cystic fluid, etc). The high flow resistance in such environments imposes strict requirements on the design features of these robots. For instance, designs of micro/nanorobots need to be able to carry out geometrically non-reciprocal motion in order to generate a net displacement in a low Reynolds number fluid environment [27, 28]. Nowadays, the researches on design strategies of micro/nanorobots focus on developing some micro/nanostructures inspired by natural microorganisms. Main types of micro/nanorobot designs such as helical [29–34], surface-assisted [35–40], ciliary [41, 42], scaffold [43–46] and biohybrid designs [47–49, 50–57] have been successfully developed to overcome flow resistance and perform tasks in a biocompatible manner. The research on fabrication techniques of magnetically actuated micro/nanorobots is an important direction in the micro/nanorobots field. The manufacturing methods and assembly techniques of traditional macroscale robots are not suitable for micro- and nanoscale robots. Therefore, considerable efforts have been made in developing new fabrication methods and assembly processes of magnetically actuated micro/nanorobots.

In this review article, we systematically summarize the recent advances in design, fabrication and application of magnetically actuated micro/nanorobots, as illustrated in figure 1. Recent progresses of various robot designs, including helical, surface-assisted, ciliary, scaffold and biohybrid robots, are discussed firstly, and some representative magnetically actuated micro/nanorobots are summarized in table 1. Then, we introduce the main fabrication methods of magnetically actuated micro/nanorobots and their application attempts in targeted drug delivery, minimally invasive surgery and cell manipulation. Finally, we make a summary and discuss current challenges and future work for magnetically actuated micro/nanorobots.

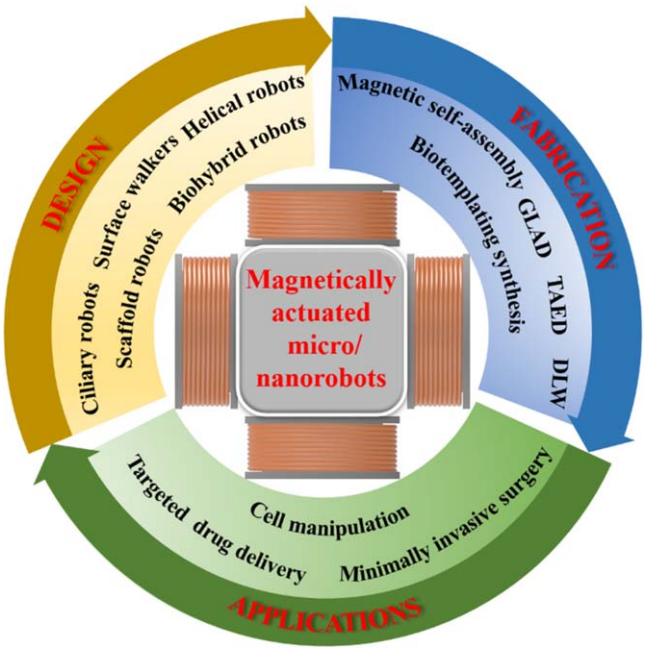

Figure 1. Schematic overview of magnetically actuated micro/nanorobots, including design, fabrication and applications.

2. Designs of magnetically actuated micro/nanorobots

2.1. Helical micro/nanorobots

The micro/nanorobots produce a net translational displacement through the nonreciprocal motion to achieve the effective propulsion in a low Reynolds number environment. For achieving such features, a design strategy inspired by the bacterial flagellum is to introduce the magnetic helical geometries on micro/nanorobots, which can be driven by exerting magnetic torques inner the body upon the application of a low-strength rotating magnetic field [72]. Typically, a helical micro/nanorobot consists of either a magnetic helical tail itself or attached to a magnetic head. They obtain a corkscrew-like motion through converting the rotational motion around its helical axis into a nonreciprocal translational movement [73, 74]. By switching the counterclockwise or clockwise direction of the external magnetic field, it is easy to achieve the forward or reverse movement of helical robots [5]. The surface hydrophobicity of helical micro/nanoswimmers plays an important role in the swimming performance,

Table 1. Representative magnetically actuated micro/nanorobots.

Synthetic materials	Design strategies	Fabrication methods	Applications	Speed (μm s^{-1})	Size (μm)	References
Photosensitive polymers, Ni, Au	Helical robots	DLW	Targeted drug delivery	62.5	15 × 5	[31]
Zinc-based MOF, zeolitic imidazole, framework-8 (ZIF-8), Ni	Helical robots	GLAD	Targeted drug delivery	50	10	[33]
PEGDA700, MNPs, ethylenediamine	Helical robots	DLW	Targeted drug delivery	160	100 × 35	[58]
Au, Ag, Ni	Surface walkers	TAED	Minimally invasive surgery	13.19	6 × 0.6	[35]
$FeCl_2.4H_2O$, $FeCl_3.6H_2O$, PDMS	Surface walkers	DLW	Targeted drug delivery	1000–5020	—	[59]
Au, Ni, SiO_2	Surface walkers	TAED	Targeted drug delivery	600	3.0 ∼ 7.8	[60]
PDMS, Fe	Ciliary robots	TAED	Targeted drug delivery	640	17 × 7	[61]
IP-L 780, Ni	Traveling-wave robots	DLW	Minimally invasive surgery	25	120 × 16	[62]
PLGA, Fe_3O_4	Scaffold robots	TAED	Cell manipulation	250	≈100 × 20	[43]
SU-8, Ni, Ti	Scaffold robots	DLW	Cell manipulation	≈1500	70–90	[44]
SiCN, MNP	Scaffold robots	DLW	Cell manipulation	≈85.56	42 × 15	[45]
RBCs, IGG, Fe_3O_4	Biohybrid robots	BTS	Targeted drug delivery	56.5	≈2	[47]
Pine pollen, Fe_3O_4	Biohybrid robots	BTS	Targeted drug delivery	175.19	25	[50]
Unicellular green microalgae *C*, magnetic polystyrene (PS)	Biohybrid robots	BTS	Targeted drug delivery	≈109.54	20	[53]
S.platensis, $BaTiO_3$, Fe_3O_4	Biohybrid robots	BTS	Cell manipulation	333.3	≈22 × 0.6	[55]
Dynabeads M-450 epoxy	Surface walkers	MSA	Targeted drug delivery	≈25	12.5	[63]
SiO_2, Ni, perfluorocarbon silane	Helical robots	GLAD	Minimally invasive surgery	0.7 – 11.4	5×10^{-2}	[64]
Fe, Pt, SiO_2	Helical robots	GLAD	Cell manipulation	24	1.5	[65]
P(VDF-TrFE), FeGa	Surface walkers	TAED	Targeted drug delivery	12	$\approx 2.5 \times 10^{-2}$	[66]
Spirulina platensis, bacteria, Fe_3O_4	Biohybrid robots	BTS	Minimally invasive surgery	21.7–78.3	≈50	[67]
IP-S photoresist, Ni, TiO_2	Scaffold robots	DLW	Cell manipulation	—	95 × 300	[68]
PEGDA, PETA, Fe_3O_4	Helical robots	DLW	Minimally invasive surgery	82	120	[69]
Ormocomp, PMMA, PDMS, Ti, Fe	Helical robots	DLW	Cell manipulation	≈800	130– 170	[70]
PM, sulphonyl esters, γ-Fe_2O_3, Pt	Surface walkers	MSA	Cell manipulation	2.0 ± 0.05	≈4.5	[71]

DLW, direct laser writing; TAED, template-assisted electrochemical deposition; GLAD, glancing angle deposition; MNPs, magnetic nanoparticles; PDMS, poly(dimethylsiloxane); IGG, indocyanine green; IP-L 780, liquid negative-tone photoresist; PLGA, poly(lactic-co-glycolic acid); SU-8, a negative photoresist; RBCs, red blood cells; BTS, biotemplating synthesis; MSA, magnetic self-assembly; P(VDF-TrFE), poly(vinylidene fluoride-trifluoroethylene); FeGa, an iron-gallium alloy; PEGDA, poly(ethylene glycol) diacrylate; PETA, pentaerythritol triacrylate; Ormocomp, a negative tone photoresist; PM, commercially available superparamagnetic microspheres.

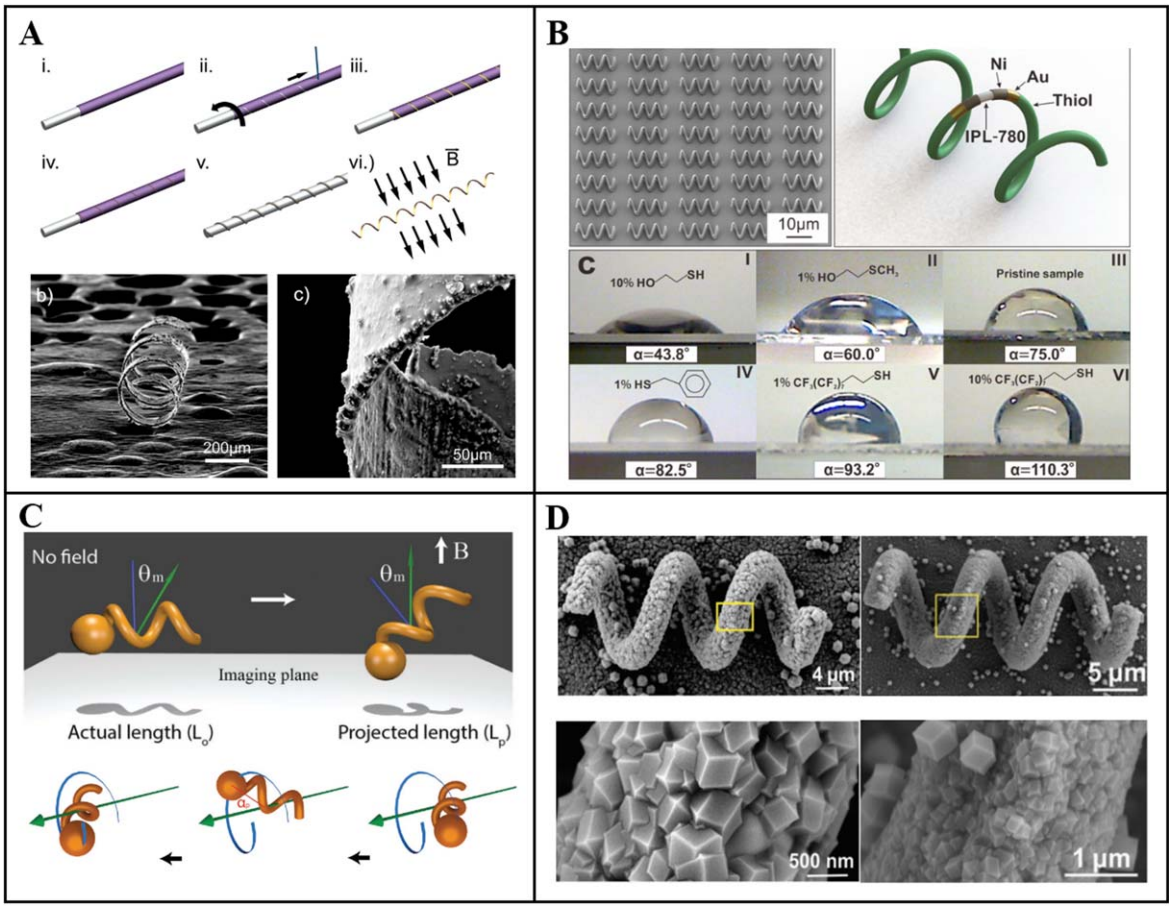

Figure 2. Helical micro/nanorobots. (A) Fabrication processes and SEM images of CoPt-coated ABFs. [30] John Wiley & Sons. © 2018 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (B) Architecture of helical microswimmers and characterizations of modified surface chemistry. Reprinted with permission from [31]. Copyright © 2018, American Chemical Society. (C) Helical micro/nanoswimmers for mobile viscometers. [32] John Wiley & Sons. © 2018 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (D) SEM images of MOF-based helical microswimmers. [33] John Wiley & Sons. © 2019 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

which has been proved by the theoretical and experimental analyses. The two helical microswimmers with different surface wettability were both actuated at low Reynolds fluid media, and experimental results demonstrated that more hydrophobicity would give rise to better propulsion performance [75, 76]. The propulsion force of helical robots mainly relies on the actuation magnetic field gradients, which is presented as macroscopic drag behavior [77, 78]. Magnetic helical micro/nanorobots have been extensively explored to date on various fundamental aspects owing to their efficient propulsion capabilities in highly viscous media, which is supported by a large number of relevant literatures.

Often referred to as artificial bacterial flagella (ABF), the bacterial flagellate-like helical micro/nanorobots are commonly used as carriers for transporting and releasing of micro/nanometer-size cargos. Qiu *et al* in 2014 developed the non-cytotoxic ABFs with the helical bodies fabricated by a commercially available ORMOCOMP photoresist. The helical microstructures with a length of 15.7 μm and diameter of 5.3 μm were coated with Fe for magnetic actuation. Under a rotating magnetic field of 9 mT at 72 Hz, the ABFs were propelled forward at 48.9 μm s^{-1} [29]. Considering the excellent hard magnetic properties of CoPt, resear0chers selected CoPt to build the helical bodies of ABFs. The preparation processes were as follows (figure 2(A)): firstly, Al wires were coated by using the isolating polymers; secondly, the isolation polymers were patterned with the laser; thirdly, the clearances were coated with the electroless gold; fourthly, the ferromagnetic materials were electrodeposited on the surface of the isolation polymer; fianlly, removing the isolation polymer and magnetizing the ABFs [30]. The surface chemistry of helical robots seriously affects the physicochemical interactions between the surface of helical bodies and fluid medium, whereas impacts the motion characteristics of robots. Wang *et al* developed the helical microswimmers whose surface chemistry can be controlled independently (figure 2(B)). The experimental and analytical results had shown that the contact angle varied nonlinearly with the step-out frequency [31]. The helical micro/nanoswimmers also are used as the mobile viscometers to measure and map the local mechanical properties (e.g. viscosity) of complex heterogeneous medium (figure 2(C)) [32]. Recently, motile metal-organic frameworks (MOFs) have started serving as the micro/nanorobot design platforms. For example, researchers had designed the MOF-based magnetic helical microswimmers (figure 2(D)) with a kind of highly biocompatible

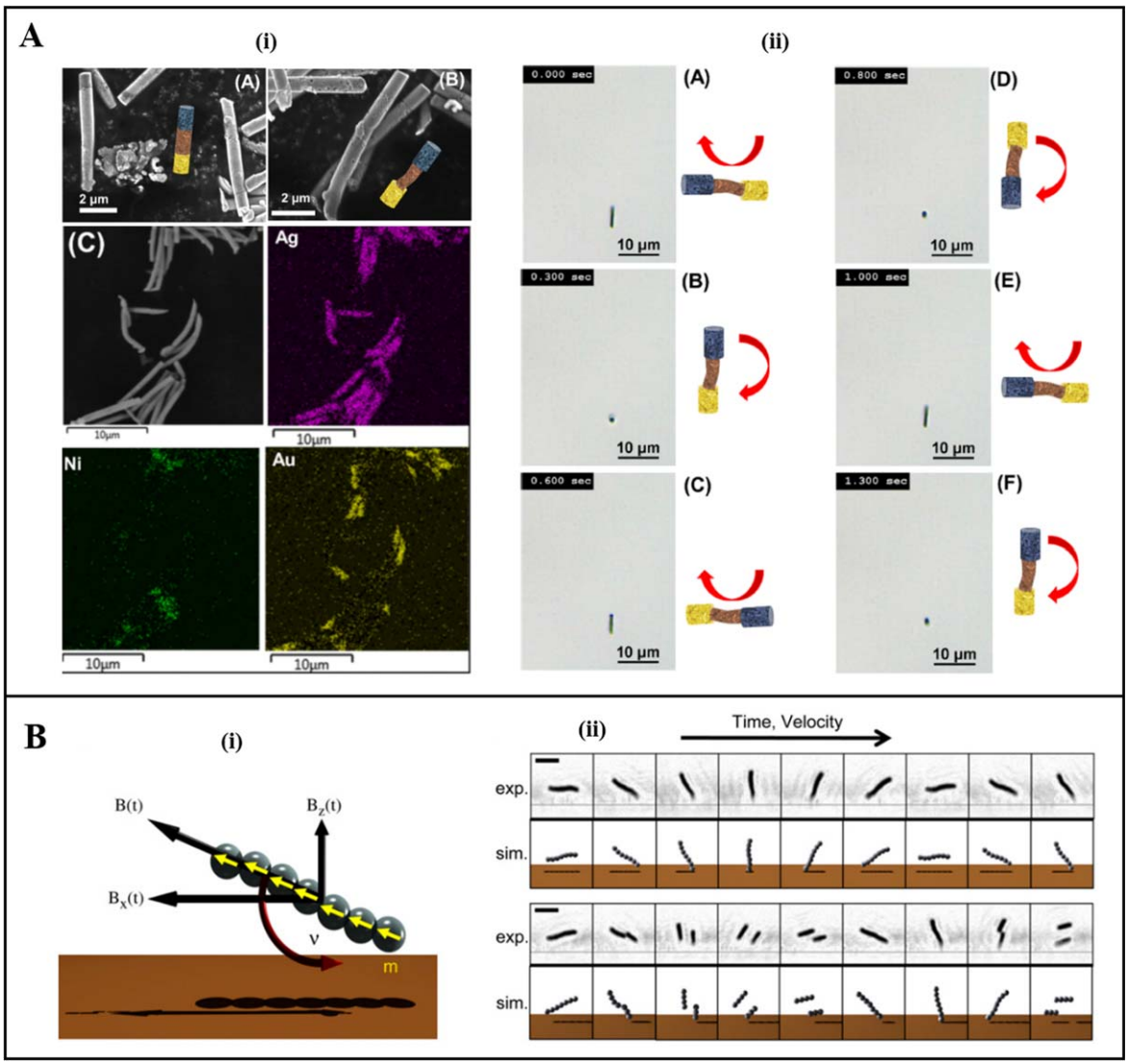

Figure 3. Surface micro/nanowalkers. (A) Au/Ag/Ni nanowires and corresponding locomotion schemes. Reprinted with permission from [35]. Copyright © 2020, American Chemical Society. (B) Self-assembled colloid microrotor chains. Reproduced with permission from [36].

and pH-responsive zinc-based MOF (ZIF-8) coating. The propelling speed of the MOF-based microswimmer increased with the frequency of the rotating magnetic field until it reached the step-out frequency (40 Hz) [33]. In order to reduce the side effects of magnetic nanoparticles (MNPs) remaining in human bodies, many efforts [34, 58, 79–82] have been made. For instance, Lee *et al* [58] reported a biocompatible and hydrolyzable PEGDA-based helical microrobot that could achieve the separation and retrieval of MNPs by using a reducing agent (DTT) and an external stimulus (NIR).

2.2. Surface micro/nanowalkers

Micro- and nanoscale surface walkers or rollers are another type of micro/nanorobots that are capable of generating a translational movement based on the 'surface-assisted locomotion' mechanism. The 'surface-assisted locomotion' mechanism refers to the passive destruction of spatial symmetry of robots by means of a physical boundary [83, 84]. In practice, surface micro/nanowalkers can achieve a walk or roll locomotion with the aid of a liquid surface, solid interface and microscale wall in a fluid or dry environment [5]. With this in mind, researchers have developed an abundance of magnetically responsive micro/nanostructures based on surface-assisted propulsion, such as micro/nanorods, micro/nanotubes, micro/nanowires, assembled colloids and Janus particles, etc [85, 86]. Vyskočil *et al* developed the Au/Ag/Ni nanowires (length of 6 μm and diameter of 600 nm) acting as the microrobotic scalpels. They had high forward propulsion efficiency and magnetic response frequencies of 70 Hz (figures 3(A)-(i)). By applying a low-strength rotating magnetic field, the microrobotic scalpels not only enter the cytoplasm of cancer cells and but also remove cytoplasmic sol while maintaining the integrity of the plasma membrane (figures 3(A)-(ii)) [35].

However, micro/nanowire-like surface walkers have usually poor cargo loading or carrying capacities. To address

this issue, self-assembled colloid surface walkers start offering precious opportunities to improve the maneuverability of cargo loading and releasing [36–39]. Sing *et al* in 2010 firstly reported the self-assembled colloid microrotors that 'walked' along a solid surface under a rotating magnetic field. These microrotors were fixed together only by magnetic forces and allowed reversible assembly and disassembly of microrotor chains (figure 3(B)) [36]. Recently, self-assembled colloid clusters have also been designed to improve carrying capacity [38, 39]. For instance, Elismaili *et al* developed the 2D paramagnetic self-assembled colloid clusters actuated by a time-varying rotating magnetic field. They studied the dynamics of self-assembled colloid clusters [40]. Based on intelligent hydrogel, Chen and collaborators proposed the child-parent microrobots (CPMs) which had four shapes of sphere, mushroom, red blood cell and teardrop. CPMs acted as a vehicle to carry child microrobots (CMs). Child microrobots were prepared with carboxymethylcellulose (CMC), Na-alginate (NaA) ferric ions and drug nanoparticles [59].

A large number of experiments and theoretical analyses have demonstrated that dynamic characteristics of self-assembled colloid surface walkers mainly rely on magnetic fields, boundary characteristics, constraint conditions and fluid properties [87–89]. Locomotion modes of magnetic surface walkers depend on the features (i.e. configuration, strength, frequency and gradient) of the external actuation magnetic field [90–93]. Xie Hui *et al* developed the reconfigurable microrobot swarm, allowing liquid, chain, vortex and ribbon-like microrobotic swarms by using a programmable alternating magnetic field that supported fast and reversible conversions among them. The chain was for narrow passages, and the herring school-like ribbon was for large areas of synchronous manipulation and the colony-like vortex was clustered at high densities to coordinate heavy loads [90]. Surface walkers based on Janus microparticles are a kind of magnetoelectric (ME) micromachines. They can be simultaneously manipulated and triggered to produce an electrical signal under a time-varying magnetic field [60, 91–94]. Chen *et al* [92] introduced the magnetoelectric Janus particle-based microrobots, which were prepared by coating SiO_2 microspheres with a bilayer composite composed of $CoFe_2O_4$ and $BaTiO_3$. Alapan *et al* also developed the cell-sized multifunctional surface microrollers with diameters of 3.0–7.8 μm. These microrollers were based on Janus microparticles prepared by sputtering gold and nickel on the surface of silica microspheres. And the movement speeds of magnetic microrollers were up to 600 μm s^{-1} [60].

2.3. Ciliary micro/nanorobots and scaffold micro/nanorobots

Magnetic micro/nanorobots based on the traveling-wave and metachronal-wave propulsion are inspired by the natural ciliated microorganisms (e.g. *Paramecia*). This type of robots breaks the temporal symmetry to produce an effective net displacement owing to the phase differences between neighboring cilia beating motion [5]. Magnetic artificial cilia [95, 96] can be designed and customized according to requirements. Various magnetic artificial cilia promote the development of magnetically actuated ciliary micro/nanorobots. Magnetic artificial cilia beating motions are affected by cilium design features (i.e. length, diameter and lip angle) and actuation magnetic fields (i.e. strength and direction). Kim *et al* in 2016 developed the magnetic ciliary microrobots with a full three-dimensional polymer body. For magnetic actuation, researchers sputtered nickel/titanium materials onto cilium structures of the microrobots. Under an electromagnetic coil system, the ciliary microrobots were propelled by eight magnetic cilium beating motions and achieved the translational motion and rotational motions [41]. The magnetic artificial cilia carpets based on the metachronal-wave propulsion have been developed while researchers have demonstrated their transport abilities and locomotion capabilities in a fluid environment and solid surface, respectively [42]. Lu *et al* developed a multilegged soft crawling robot that was capable of the metachronal-wave motion based on periodic moving magnetic fields (figure 4(A)). In their study, the movement speed of the multilegged robot was greater than 40 body length s^{-1}, the load-bearing was greater than 100 own weight, and the height of across obstacle was higher than 10 body height [61]. Moreover, multi-link or multi-segment micro/nanostructures also can produce the traveling-wave or metachronal-wave propulsion. Li *et al* in 2016 designed a magnetic artificial nanofish (length of $\sim$4.8 μm, diameter of about 200 nm) consisting of two magnetic nickel segments for magnetic actuation, two gold segments for extending the amplitude passively, and three flexible silver segments as flexible porous joints [97]. These multi-segment magnetic micro/nanorobots are powered by an oscillating magnetic field. Similarly, the work in [62] reported a magnetic microswimmer with an overall dimension of 120 $\times$ 16 μm^2 that was constructed with four rigid segments (bodies) connected by joints (figure 4(B)). In this study, the nickel (Ni) was deposited on the head of the microswimmer for magnetic actuation. The middle U-shaped structure was designed to transfer the oscillations from the head segment to the rear segment of the microswimmer for obtaining traveling-wave propulsion.

Inspired by the porous extracellular matrix, three-dimensional microscaffolds with interconnecting pores have been developed as carriers for cell and tissue culture [98–100]. Magnetically actuated micro/nanorobots based on microscaffold structures possess a large storage space and loading capacity. They effectively enhance the therapeutic efficacy of targeted drug delivery, minimally invasive surgery and cell manipulation [101–103]. Go *et al* in 2017 firstly proposed the magnetically actuated microscaffolds using for carrying mesenchymal stem cells (MSCs) to repair the articular cartilage. In this study, the 3D microscaffolds were microspheres with interconnected micro holes that provided sufficient space for cell adhesion, support and transportation (figure 4(C)). These microscaffolds were composed of the biocompatible and biodegradable poly(lactic-co-glycolic acid) (PLGA) and polyethylenimine (PEI) with Fe_3O_4 nanoparticles coating for magnetic actuation [43]. Magnetic porous spherical microscaffolds for applications of carrying and delivering targeted cells have attracted the attention of most

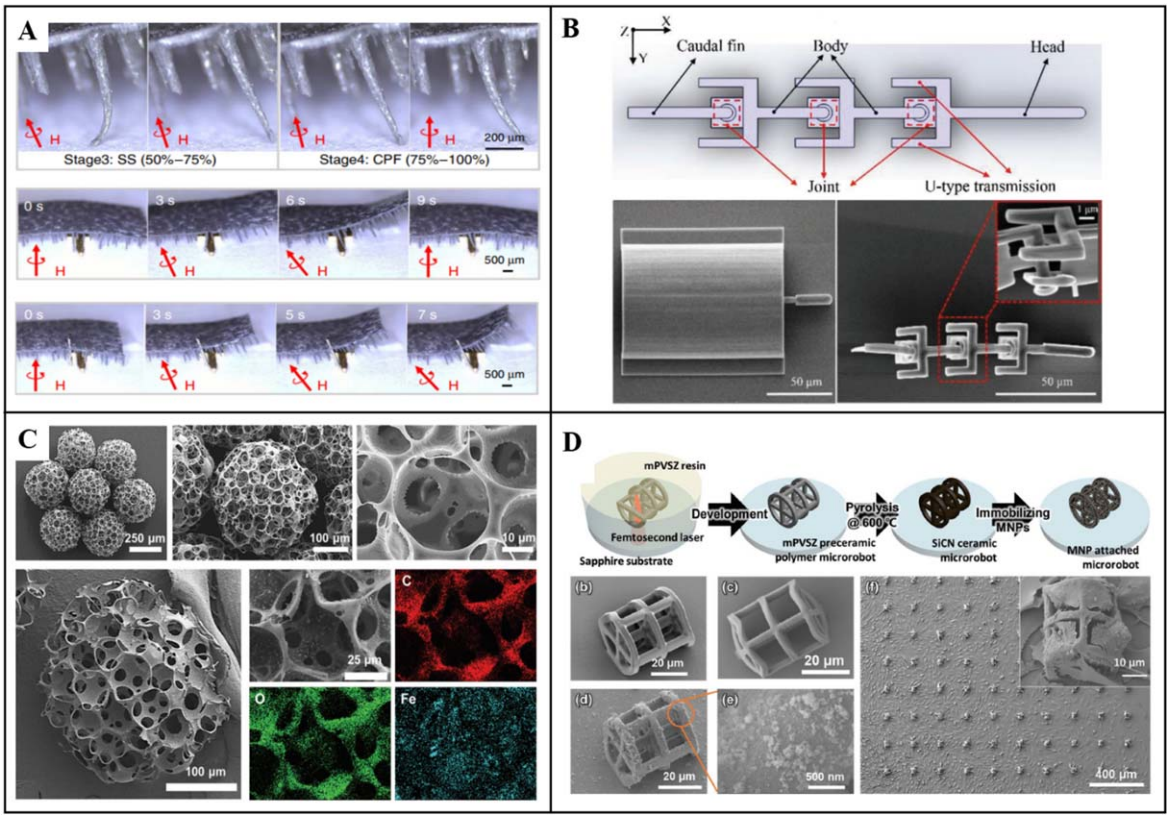

Figure 4. Ciliary micro/nanorobots and scaffold micro/nanorobots. (A) Multilegged soft millirobots performing metachronal-wave motion. Reproduced from [61]. CC BY 4.0. (B) Design scheme and SEM images of artificial undulatory microswimmer. [62] John Wiley & Sons. © 2019 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (C) SEM images of spherically porous PLGA-based microscaffolds. [43] John Wiley & Sons. © 2017 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (D) SEM images of cylindrical microscaffold structures. [45] John Wiley & Sons. © 2019 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

researchers owing to their excellent carrying capacities. Li *et al* fabricated the porous spherical microscaffold carriers via 3D laser lithography to carry and deliver the targeted cells. Similarly, these 3D microscaffolds carriers were also coated with magnetic materials (Ni) for magnetic actuation and biocompatible materials (Ti) for reducing toxicity [44]. Apart from spherical microscaffold structures, microscaffolds of other shapes have also been designed. For example, by using advanced 3D printing techniques, Jeon and collaborators have developed many microscaffold structures with complex morphologies and uniform pore sizes, such as cylinders, cuboids and spirochetes, as shown in figure 4(D) [45, 46].

2.4. Biohybrid micro/nanorobots

Biohybrid robots, also known as biosynthetic robots, are a new kind of micro/nanorobots synthetized by the living biological systems (e.g. cells and microorganisms) and non-living microelectromechanical systems (e.g. artificial nanostructures and nanoparticles) [104–107]. Biohybrid robots are made up of two components: the first refers to the living cells or microorganisms retaining ideal biocompatibility and deformability. Another refers to the artificial micro/nanostructures or nanoparticles acting as carriers for supporting these cells [108, 109]. Attributing to their excellent attributes of biocompatibility, deformability and low or nontoxicity, biohybrid robots have attracted a lot of interest from scientists in the field of robotics and biomedicine, and have developed rapidly in recent years [110–112]. At present, the designs of magnetic biohybrid micro/nanorobots are mainly based on animal/plant cells and microalgae. Therefore, the research progresses of cell-based and microalgae-based micro/nanorobots are introduced respectively in this section.

A large number of impressive cell-based micro/nanorobots have been developed to apply in biomedical systems due to excellent biocompatibility, deformability and carrying capacity of cells [101]. The representative one is the magnetic micro/nanorobotics based on red blood cells (RBCs) because RBCs have a long circulation time (∼120 d) and large space volume for carrying cargoes [113–115]. Gao *et al* developed red blood cell-mimicking micromotors (RBCMs) which were powered by an acoustic field and navigated by a magnetic field (figure 5(A)) [47]. In their study, the photosensitizers (PSs) encapsulated with magnetic hemoglobin cores were cloaked with natural RBC membrane to prepare the RBCMs using for photodynamic therapy. Yasa *et al* [116] used mouse macrophages and splenocytes to engulf the magnetic double-helical microswimmers. It was found that macrophages and spleen cells could recognize and induce different immune responses to helix turn numbers of microswimmers. Based on phagocytosis of mouse macrophages, Nguyen *et al* developed an anticancer drug delivery system that was prepared by

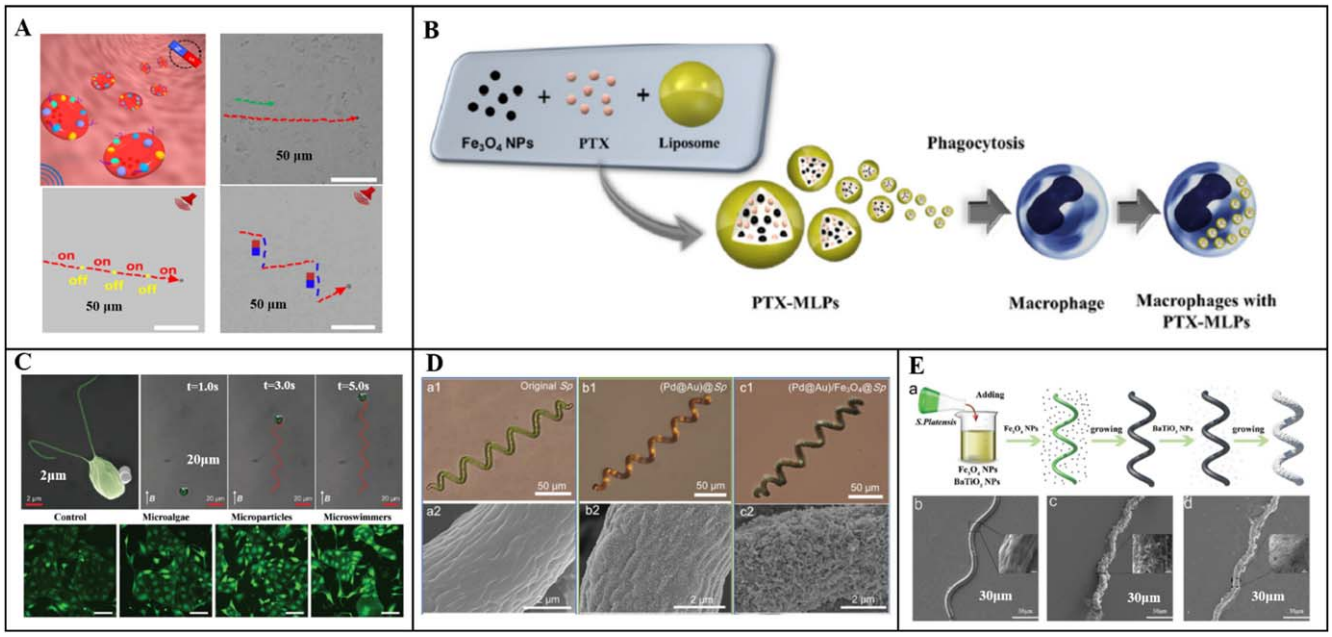

Figure 5. Biohybrid micro/nanorobots. (A) Acoustically powered RBCM micromotors are navigated by an external magnetic field. Reprinted with permission from [47]. Copyright © 2019, American Chemical Society. (B) Preparation and transport characterization of the macrophages with PTX-MLPs, where PTX-MLPs stands for (Paclitaxel, PTX)-encapsulated magnetic liposomes. Reprinted from [48], © 2016 Elsevier B.V. All rights reserved. (C) Active drugs transportation of algal microswimmers. [53] John Wiley & Sons. © 2018 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (D) *S.platensis*-based microrobot and its SEM images. Reprinted with permission from [54]. Copyright © 2019, American Chemical Society. (E) Fabrication and SEM images of magnetic *S.platensis*@Fe_3O_4@$tBaTiO_3$ micromotors. [55] John Wiley & Sons. © 2020 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

encapsulating the anticancer drugs with magnetic liposomes and mouse macrophages (figure 5(B)). The average velocity of the macrophages carrying the anticancer drugs can reach 10.48 ± 4.43 μm s^{-1} under an external magnetic field [48]. Sperm cells are capable of protecting hydrophilic drugs from dilution, immune response or biodegradation by tissue fluids in the body [117–119]. So, sperm cells also are widely used in the synthesis of magnetic cell-based micro/nanorobots. Magdanz *et al* in 2020 developed the magnetically actuated biohybrid microrobots based on nonmotile sperm cells acting as a template. In their report, researchers utilized the opposite surface charges of nonmotile sperm cells and maghemite nanoparticles to prepare biohybrid microrobots named IRONSperms by using the electrostatic interaction [49]. The experimental results showed that IRONSperms were capable of achieving the helical flagella-like propulsion in a low Reynolds numbers fluid environment upon the application of periodically varying magnetic fields.

In addition to animal cells, some plant cells, such as pollen and spore, can also be involved in the preparation of biohybrid robots. Apart from good structural uniformity, permeability and biocompatibility, the biggest advantage of pollen is that it has a large pollen cavity that can be used to carry more cargos [120–122]. Sun *et al* in 2019 designed a magnetic biohybrid micromotor based on pine pollen for the targeted drug delivery. In their work, natural pine pollen was defatted firstly to increase the permeability for compound loading ultimately, and subsequently the defatted pollen, magnetic nanoparticles and drugs were mixed evenly and vacuumized [50]. Moreover, spores also can be used to synthesize the magnetically actuated biohybrid microrobots [51, 52]. For instance, Zhang *et al* directly deposited magnetic nanoparticles and then encapsulated functionalized carbon nanodots on porous natural spores to synthesize magnetic spore-based microrobots. The fluorescent magnetic spore@Fe_3O_4@CDs microrobots were used to detect toxins secreted by Clostridium difficile (*C. diff*) [52].

Researchers have found that microalgaes possess the excellent properties of highly efficient helical propulsion. Therefore, some magnetic microalgae-based biohybrid micro/naonorobots have been successfully developed for active cargos delivery. Yasa *et al* [53] attached microalgae (*C. reinhardtii*) to the surface of polyelectrolyte (PE) functionalized with magnetic polystyrene (PS) microparticles to prepare the magnetic microalgae-based microswimmers for cargos delivery. The non-covalent electrostatic interaction between positively-functionalized magnetic PS particles and negatively charged microorganisms was utilized to attach the cargos noninvasively to the microalgae without affecting their natural movement (figure 5(C)). Wang *et al* in 2019 [54] developed the magnetic *Spirulina* (*Sp.*) biohybrid microrobots for targeted drugs delivery and synergistic chemophotothermal therapy. In their work, Pd@Au nanoparticles (NPs) were firstly deposited in the *Sp.* bio-templates acting as photothermal conversion agents, and then the Fe_3O_4 NPs were deposited in the prepared (Pd@Au)@*Sp.* to obtain the (Pd@Au)/Fe_3O_4@*Sp.* microrobots. Experimental results showed that the highest movement speed of (Pd@Au)/Fe_3O_4@*Sp.*-DOX microrobots was up to 526.2 μm s^{-1} upon application of a rotating magnetic field (figure 5(D)). By coating *Spirulina platensis* (*S.platensis*) as a

bio-template with Fe_3O_4 nanoparticles for magnetic actuation and $BaTiO_3$ nanoparticles working as sensitive materials, Liu and collaborators developed the *S.platensis*@Fe_3O_4@t$BaTiO_3$ micromotors using for precise neural cell electrical stimulation (figure 5(E)) [55]. In addition, many research groups have also developed some *Spirulina*-based magnetic biohybrid micro/nanorobots for targeted drugs delivery/treatment [56, 57].

3. Fabrication techniques of magnetically actuated micro/nanorobots

At present, mature CNC machines have realized the high-precision machining of macro-scale robots. However, when robots are sized down to microns or nanometers, CNC machines are no longer suitable [1]. Therefore, it is urgent to develop new micro/nano manufacturing technologies suitable for micro/nanorobots. Other review articles [123] have discussed in detail the advances in preparation of magnetic nanoparticles. This section focuses on the main progresses in manufacturing techniques of magnetically actuated micro/nanorobots. Up to now, the common fabrication methods of magnetically actuated micro/nanorobots include magnetic self-assembly (MSA), glancing angle deposition (GLAD), template-assisted electrochemical deposition (TAED), bio-templating synthesis (BTS) and direct laser writing (DLW).

3.1. Magnetic self-assembly (MSA)

Apart from operating magnetically powered microscopic robots as individual units, active magnetic particles with collective or swarm behaviors have been recently investigated for on-demand *in vivo* applications [124]. Taking inspiration from natural phenomena of insect swarms [125] or birds flocks [126], such self-assembly abilities allow to accomplish more complicated tasks for localized transportation [63] and targeted cargos releasing [127] in a precisely rapid and programmable manner. The common pattern of these active materials is observed to be out-of-equilibrium, which can be stimulated and actuated by physical or chemical means (catalytic conversion or self-electrophoresis [128]) or external fields (optical, magnetic or electric [129]). The magnetic field enables to provide designated and long-range forces deeply penetrating tissues in a harmless fashion, without having continually interactions among pathway media [130].

As mentioned above, magnetic self-assembly, owing to its versatile multifunctionalities by controllable structure reconfiguration, has been regarded as an effective alternative to fabricate swarm robots. Taking advantage of the 2D flat plane rotating magnetic field, Tasci *et al* [131] developed the superparamagnetic beads capable of self-assembling into size-controlled microwheels. However, field-induced magnetic torques can only make these lying microwheels spin *in situ* without net translational displacement. After applying an additional normal component to the current magnetic field, the microwheels could eventually be propelled and reoriented in a predefined camber angle at a speed of around 100 μm s^{-1} (figure 6(A)). In order to load and unload cargos under a harsh fluid environment, Yang *et al* [63] initially created superparamagnetic linear-linked colloidal chains by applying a 1D magnetic field of around 20 mT in the vertical direction of a diluted epoxy-functionalized Dynabeads solution, and then another 2D circularly rotating magnetic field generated by Helmholtz coils was used for transforming chain into lassos (figures 6(B)–(i)). By tunning the external field strength ratio and phase lag of additional orthogonal AC magnetic field, the microlassos can achieve precisely spatial cargo delivery, including capture, transport and site-specific release (figure 6(B)–(ii)). Besides, superparamagnetic beads with self-assembled morphology can also migrate with programmable boundaries in magnetoacoustic composite field [132].

Except for commonly used colloidal particles, Wang *et al* [130] studied magnetic droplets consisting of benzyl-ether based water suspension and carbonyl iron microparticles. These ferromagnetic particles settled to the droplet bottom in the presence of a gravity field (figure 6(C)–(i)), and would be rapidly self-assembled into a chain once applying a processing field. Meanwhile, the magnetic droplets could realize automated navigation through modulating external field gradients to present controllable dynamic patterns (figure 6(C)–(ii)). Similarly, Fan *et al* [133] demonstrated that liquid microrobots based on ferrofluid droplets even could pass a certain channel that was narrower than their own diameter by elongating themself, that is, the various deformable or reconfigurable morphologies endowed liquid-related microrobots with unexpected practical properties (figure 6(D)). Other than multi-dimensional rotating fields, the developed ribbon-like swarm robots containing paramagnetic Fe_3O_4 nanoparticles not only showed flexible reconfigurability but also reversible scalability under a programmed oscillating field [134] (figure 6(E)).

3.2. Glancing angle deposition (GLAD)

Physical vapor deposition (PVD) constructs micro/nanostructures by vaporizing the target materials and then depositing them to a substrate [135, 136]. PVD offers a miniaturized fabrication tool for constructing highly sophisticated structures [137]. Glancing angle deposition (GLAD) is an extension of conventional PVD-based methods in which the substrate is synchronously manipulated with incident vapor deposition, rather than typically performed on a fixed substrate. Specifically, GLAD-based production procedures are divided into the following four primary steps (figure 7(A)): (i) oblique deposition of core nuclei; (ii) nuclei grow casting shadows; (iii) columns develop suppressing the shadowed growth; (iv) columns grow at a certain tilt angle [138].

Therefore, only by rotating the substrate at a constant rate after fixing the initial seed layer, vertical posts can be broadened into helical morphologies, which is considered as a feasible way to fabricate helical magnetic micro/nanorobots [32, 139, 140]. Based on this approach, Walker *et al* [139] took 500 nm silica beads spreading on the prepared silicon wafer to create a seeds monolayer as the nucleation site, and then deposited nickel (Ni) flux onto silica particles randomly at an incident angle of 85° for magnetic actuation. Similarly,

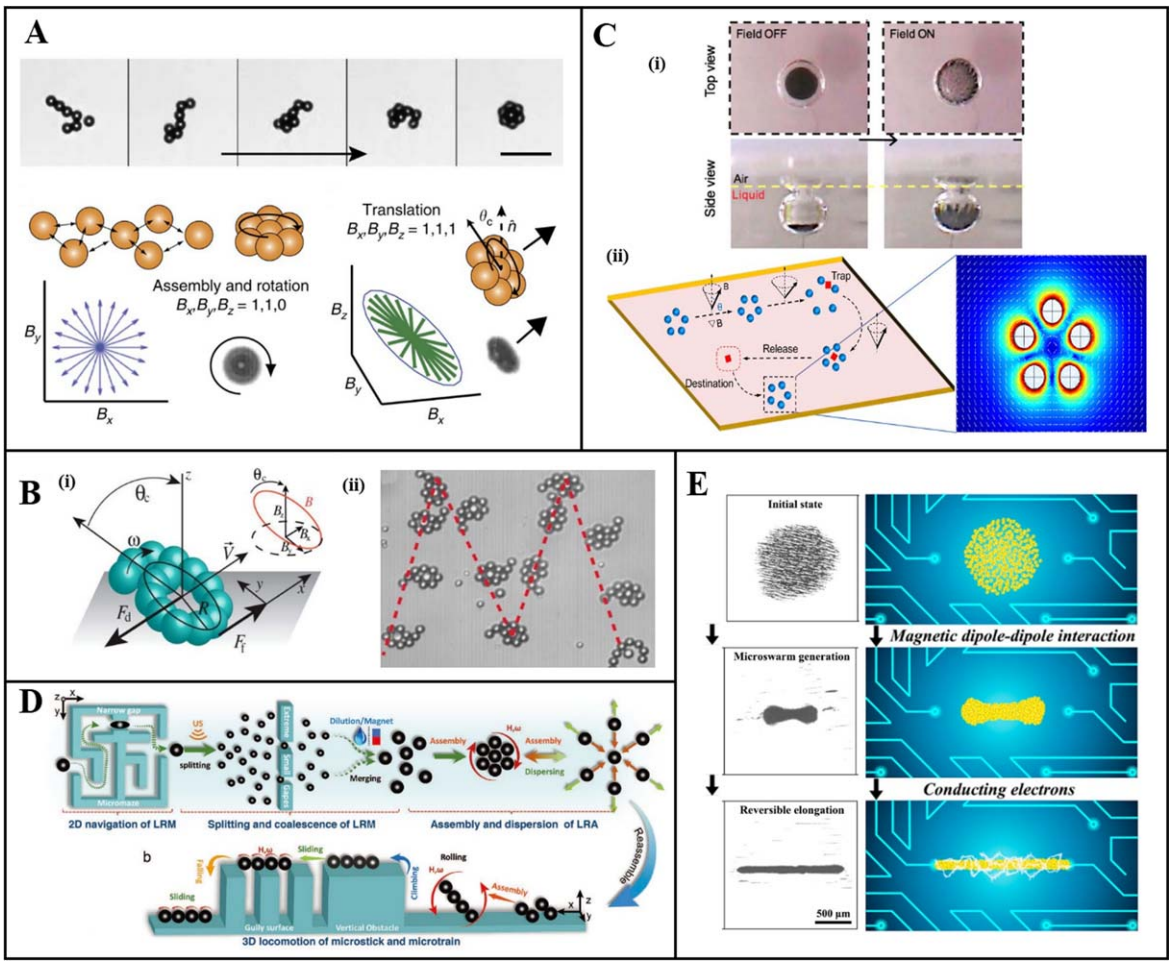

Figure 6. Magnetic self-assembly. (A) Planar rotating magnetic field combining with additional normal field component enables beads-based microwheels to generate a net translational motion. Reproduced from [131]. CC BY 4.0. (B) Magnetically actuated microlassos for cargos transportation. Reprinted with permission from [63]. Copyright © 2017, American Chemical Society. (C) Magnetic droplets reconfiguration via dynamically self-assembled motion. Reprinted with permission from [130]. Copyright © 2019, American Chemical Society. (D) Morphological transformations between liquid-microrobot monomer (LRM) and liquid-robot aggregate (LRA). [133] John Wiley & Sons. © 2020 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (E) The microswarm generation and its reversible elongation for electrical connection under an oscillating magnetic field. Reprinted with permission from [134]. Copyright © 2019, American Chemical Society.

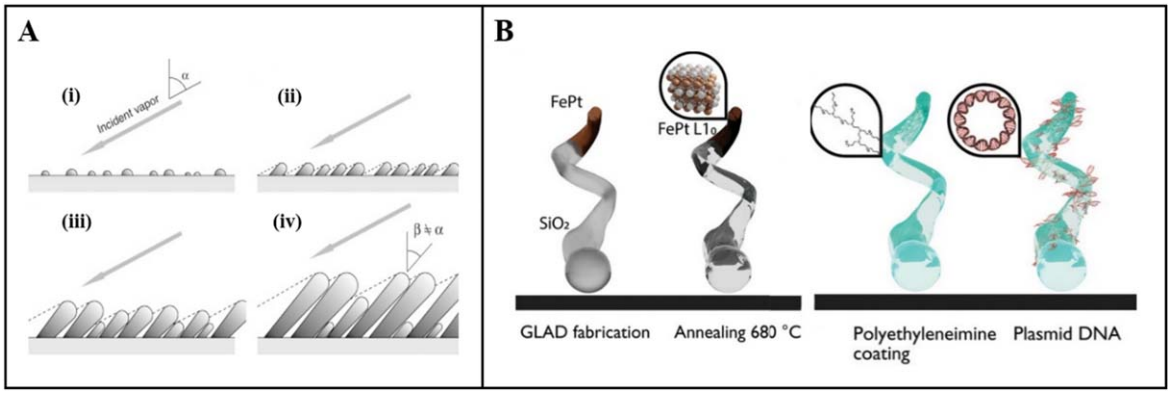

Figure 7. Glancing angle deposition (GLAD). (A) Conceptual schematic of GLAD growth. Reprinted from [138], Copyright © 2010 Peter M. Martin. Published by Elsevier Inc. All rights reserved. (B) Fabrication representation of $L1_0$ FePt nanomotors. Reproduced from [65]. CC BY 4.0.

Wu *et al* [64] prepared the left-handed SiO_2-Ni microhelices with 1.8 turns by adjusting the direction and speed of the substrate, and then the tail magnetization was continued for further magnetic actuation experiments. In contrast to the previous random distribution of seeds on a wafer substrate, Venkataramanababu *et al* [141] manufactured a 2D array of photoresist pillars to form a patterned wafer by utilizing laser interface lithography prior to the next deposition of a seed layer for higher shape customization of helical robots. Although this method makes it possible for massproducing uniform helical microstructures, material selection on either constructing helices or post deposition still hinder the properties enhancement and capabilities improvement. Thanks to the chemical inertness and biocompatibility of FePt, magnetic sections were deposited with FePt to make them more suitable for biomedical applications (figure 7(B)) [65], which compared with other magnetic materials, for example, nickel [139] was known to toxic, and iron [142] may cause stress to cells.

3.3. Template-assisted electrochemical deposition (TAED)

Electrochemical deposition is a simple but useful approach to form an adherent coating layer of desired materials onto a prefabricated conductive template or substrate. Electrolysis of liquid mixtures produces the dispersed charged particles [143]. Because of the low-cost, rapid and easy operation of this deposition technique, it is regarded as a suitable method to mass production of micro/nanostructures. Various microstructures have been deposited with the assistance of well-designed templates, including helical [144], tubular [145], rod-like [146] and etc. Anodic aluminum oxide (AAO)-assisted electrochemical deposition is one of the most popular template-assisted electrochemical deposition (TAED) strategies. This method is mainly used to fabricate nanowires with desired architectures. In general, both sides of the AAO porous template are metalized as working electrodes to facilitate the next oxidation-reductions. The AAO template is dissolved after depositing nanowires. In addition, the length of the above-mentioned nanostructures depends on the electrodeposition time. The completed prototypes need to be further magnetized to saturation in order to drive these micro robots (e.g. wire-like [66] or helical [30]) under an external magnetic field. However, once the applied magnetic source removal occurs, remanent magnetization of ferro- and ferrimagnetic nanowires would aggregate themselves together, which seems to be an austere challenge for biomedical applications because it may cause vascular blockage or even more serious diseases. For this reason, Jordi Sort *et al* [147] proposed a tri-segmented strategy (i.e. hard- and soft-ferromagnetic units separated by a copper spacer) that embedded CoPt/Cu/Ni into polycarbonate nanorods via electrochemical deposition (figure 8(A)). Based on this multi-phase method, Chen *et al* [66] reported the biphasic FeGa@P(VDF-TrFE) tubular nanostructures, in which the patterned surface can be maneuvered on procession steering and cargo release by regulating magnetic field from rotating to alternating. The designed actuation system was capable of providing diverse magnetic sources that enable drug transportation more effective while inhibiting significantly the side effects (figure 8(B)). Moreover, researchers introduced semihard magnetic materials (i.e. CoPt) to coat nanowires based on AAO-template demolding and premagnetization process (figure 8(C)), the proposed nanorobots can perform tumbling, procession and rolling on different rotating frequencies of a rotating magnetic field [89]. When assembling several individual segments with hinges (e.g. Ag), the deposited flexible body could endow micro/nanorobots with the additional degrees of freedom. Au–Ag–Ni–Ag–Au soft nanorobots inspired by fluidic trapping were able to perform 6 motion modes under the conditions of an oscillating field. They had better accomplished tasks of *in vivo* transportation and even cells delivery as included (figure 8(D)) [148]. Similarly, a multilinked two-arm nanorobot consisting of Ni–Ag–Au–Ag–Ni could execute out-of-phase wobbling, and even get a maximum velocity of $\sim$59.6 μm s^{-1} (figure 8(E)) [149].

Furthermore, TAED technique can also be employed to produce helical micro/nanorobots. The aforesaid magnetic self-assembly and GLAD are capable of fabricating helices, but almost less than 2 turns, and the entire structure seems to be unstable as accumulating to long-range length associated with the limitations of the selected materials and fabrication methods. Instead, it is possible for three-dimensional TAED to construct relatively complex architectures as long as providing corresponding templates [150–152]. Based on this approach, Li *et al* [150] took the pre-designed polycarbonate membranes with a 400 nm pore size as a template. They dissolved in methylene chloride for leaving only already deposited helical nanostructures. Afterwards, the current surface would be modified by sputtering a gold layer that produces negative charge for final platelet-membrane-derived vesicles tight adhesion promoting biocompatibility (figure 8(F)).

3.4. Biotemplating synthesis (BTS)

Although the above fabrication methods provide some suitable solutions for most common applications, it is still extremely challenging for micro/nanorobots to achieve ideal deformability and biocompatibility *in vivo* scenarios. Microorganisms in nature not only have various fine microstructures, but also have good biodegradation, self-fluorescence and magnetotactic properties. Recent studies have shown that biohybrid micro/nanorobots that are prepared by combining appropriate microorganisms with artificial micro/nanomaterials retain their excellent biological properties when they migrate in low Reynolds number fluids. Thus, such biotemplating synthesis method empowers micro/nanorobots with therapeutic capabilities for a wide range of biomedical applications in a harmless and biocompatible manner. In practice, similar attempts have been made on the basis of micro/nanorobotics among the state-of-art literature, for instance, by combining with microalgae, blood cells and bacteria [153, 154]. The natural microalgae with a remarkable feature of 3D helical micro- and nanoarchitecture, *Spirulina* (*Sp.*), is frequently taken as biotemplates to construct the

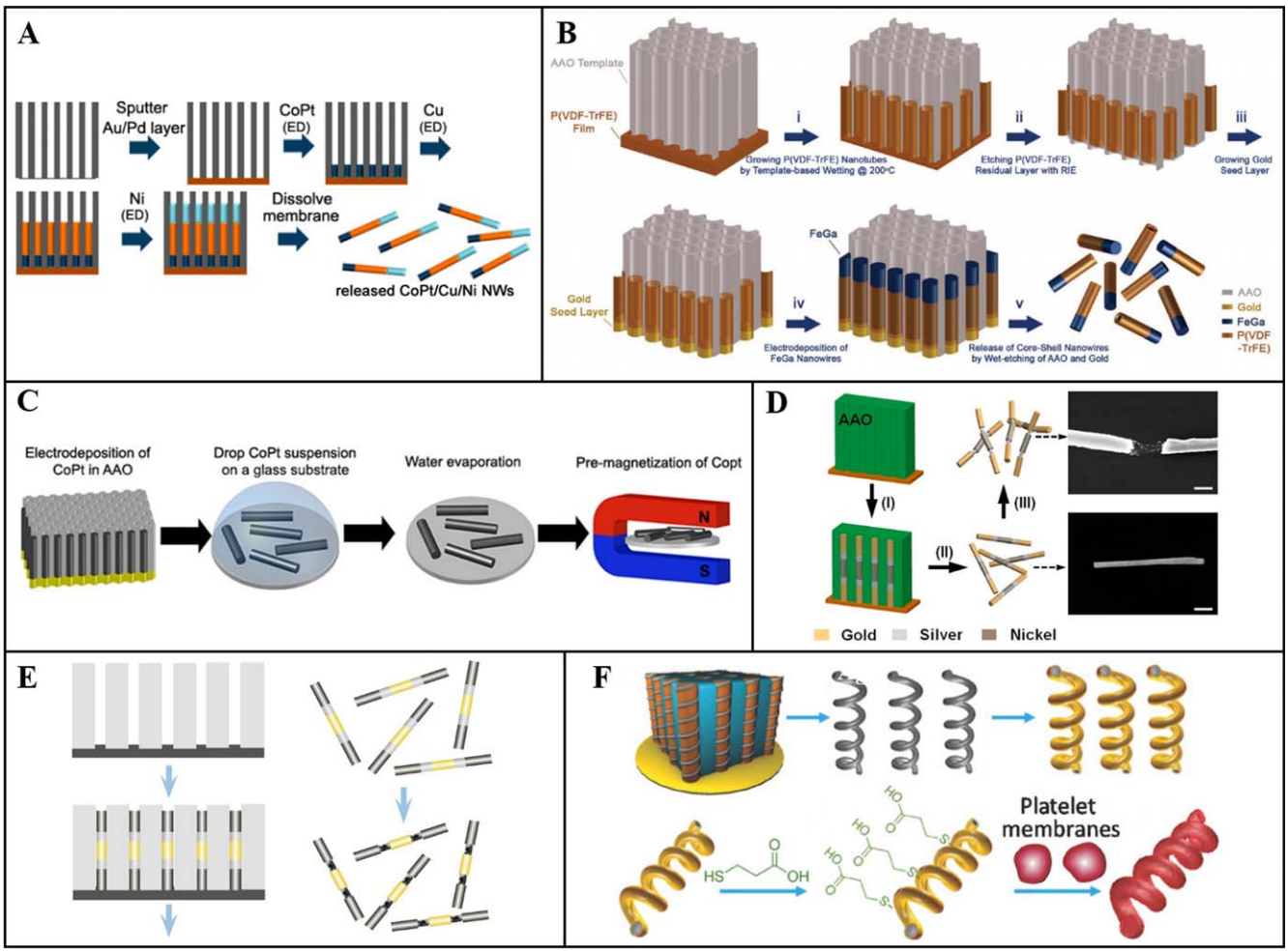

Figure 8. Template-assisted electrochemical deposition (TAED). (A) Fabrication procedures of CoPt/Cu/Ni nanowires. Reprinted with permission from [147]. Copyright © 2016, American Chemical Society. (B) Fabrication procedures of FeGa@P(VDF-TrFE) core–shell nanowires. [66] John Wiley & Sons. © 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (C) Synthesis process of CoPt nanowires. Reprinted with permission from [89]. Copyright © 2019, American Chemical Society. (D) Synthesis process and EDX mapping of swinging flexible nanomotors. Reprinted with permission from [148]. Copyright © 2021, American Chemical Society. (E) Fabrication process of two-armed nanoswimmers. Reprinted with permission from [149]. Copyright © 2017, American Chemical Society. (F) Preparation process of the platelet-membrane-cloaked magnetic helical nanomotors. [150] John Wiley & Sons. © 2017 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

biohybrid helical microrobots for cargo delivery or water decontamination, etc [57,155–157]. Such microalgae-based magnetic microrobots recently developed by Wang *et al* [54] dramatically improved the swimming speed to 526.2 μm s^{-1} under a rotating magnetic field (figure 9(A)). Similar to TAED, the fabrication process can be broadly described as sequential deposition of different functional micro/nano-particles. More specifically, Pd@Au nanoparticles are firstly embedded into pure *Sp.* cell scaffolds by electroless plating for improving photothermal conversion efficacy, then Fe_3O_4 nanoparticles would be deposited via the sol-gel method to obtain (Pd@Au)/Fe_3O_4@*Sp.* microrobots actuated by a rotating magnetic field. Importantly, photothermal conversion can be activated by applying the near-infrared laser irradiation and the subsequent temperature elevation would occur on cargo-loaded microrobots, so as to accelerate the degradation process for targeted cargo release. Apart from cargo delivery, solely synthesizing Fe_3O_4 with *Sp.* cells via a dip-coating process are also capable of performing excellent tumor inhibition under laser irradiation, by generating oxygen and then releasing chlorophyll for alleviating tumor hypoxia merely through inherent photosynthesis and organism itself (figure 9(B)) [67]. In a similar fashion, fluorescence, anther natural characteristic of *Sp.* cells, endows these micro-swimmers with the capability for accurately tracking *in vivo* motion (figure 9(C)) [56].

Red blood cells (RBC) are another potential candidate for building cell-based microrobots because they lack a nucleus, offering a more spacious space, and hemoglobin carry oxygen [158]. Gao *et al* [47] emphasized that the double-concave structure of RBC enabled it to navigate more accurately with the help of an external magnetic field and can accommodate more photosensitizers (figure 9(D)). In contrast, sperm cells not only have the ability of self-propelling but also isolate hydrophilic cargos in case of degradation or dilution from body fluids [117, 159]. A representative example reported by

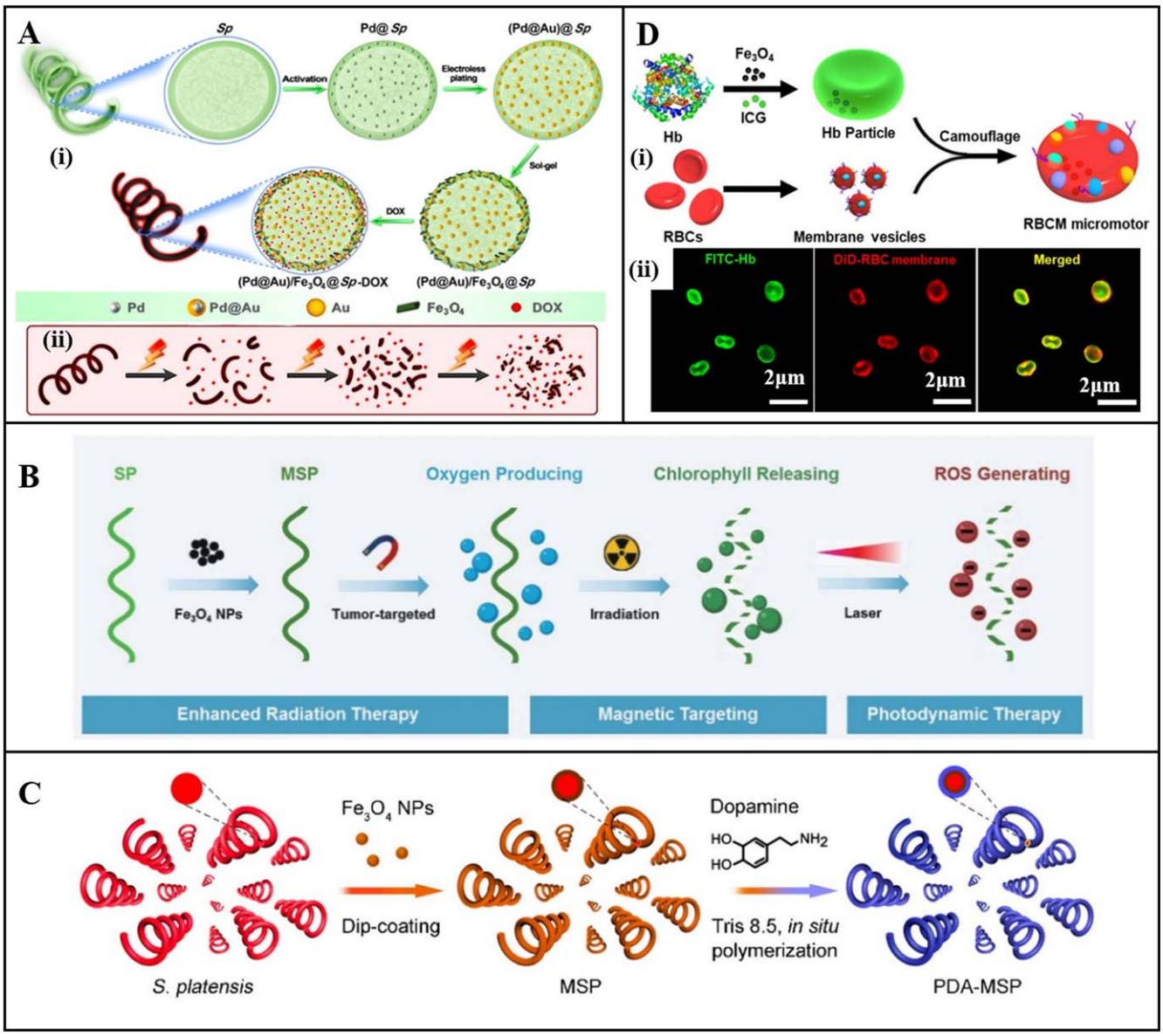

Figure 9. Biotemplating synthesis (BTS). (A) Fabrication scheme of (Pd@Au)/Fe_3O_4@*Sp*.-DOX microrobots and near-infrared triggered degradation for drug release. Reprinted with permission from [54]. Copyright © 2019, American Chemical Society. (B) *Sp*.-based microswimmers for enhancing radiation therapy, magnetic tumor targeting and photodynamic therapy. [67] John Wiley & Sons. © 2020 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (C) Fabrication procedures of polydopamine (PDA)-coated magnetic *Sp.* cells. Reprinted with permission from [56]. Copyright © 2020, American Chemical Society. (D) Fabrication and CLSM images of red blood cell-mimicking micromotors. Reprinted with permission from [47]. Copyright © 2019, American Chemical Society.

Sarthak *et al* was IRONSperms fabricating with maghemite nanoparticles [49]. The embedded metal-based nanoparticles enabled IRONSperms swarms to move along the travelling-wave generated by an applied rotating field while can be located via ultrasound feedback.

3.5. Direct laser writing (DLW)

For fabrication of highly deformable, biocompatible and biodegradable micro/nanorobots, biotemplating synthesis technique has garnered extensive attention in extending trains of thought of traditional micro/nano manufacturing techniques while addressing more intractable application-determined problems. However, magnetic nanoparticles tend to weaken the mechanical strength of biotemplates, and also may present uncontrollable aggregation after biotemplates decomposition. For fabrication methods without introducing predetermined templates, such as GLAD and magnetic self-assembly, highly sophisticated peripherals and control systems are usually required to ensure the process resolution and product accuracy. TAED presents in terms of fewer operational issues and structural defects because of fewer manual intervention during fabrication, but it only manufactures some simple architectures (e.g. rod or wire-like). Direct laser writing (DLW) based on two-photon polymerization (TPP) nanolithography is capable of manufacturing arbitrary 3D

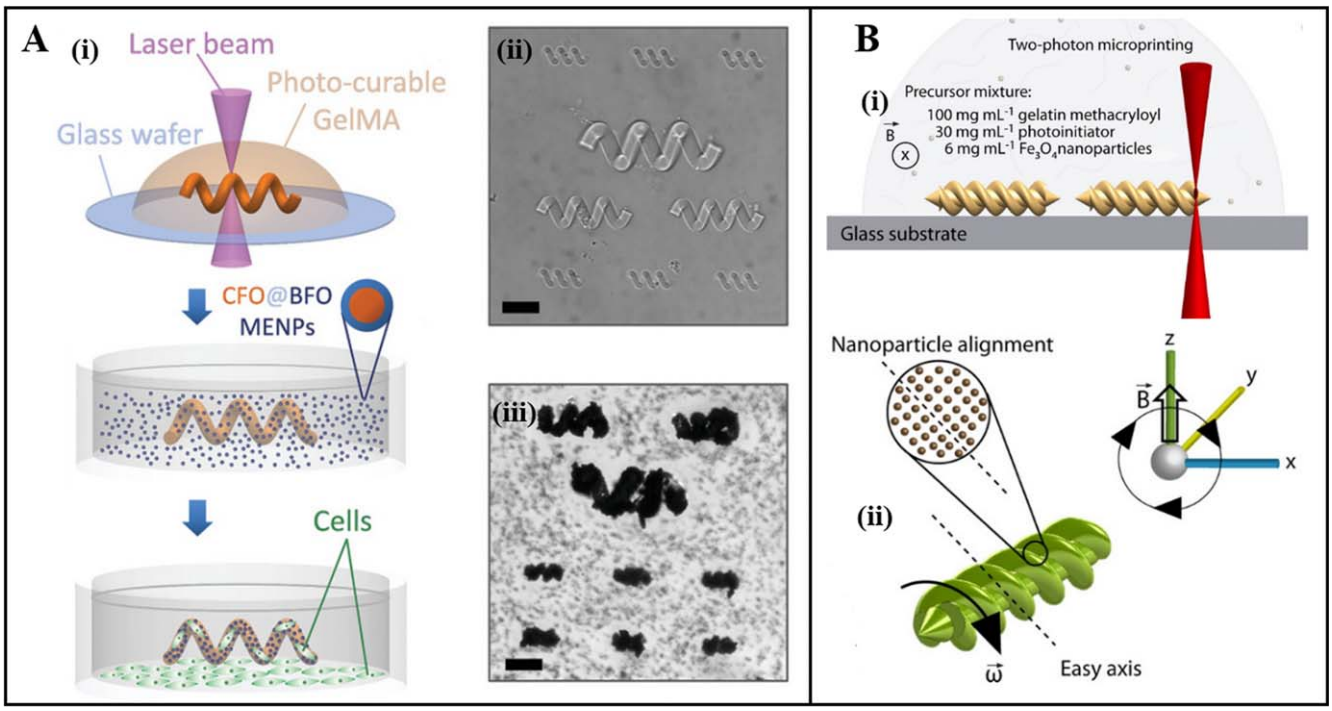

Figure 10. Direct laser writing (DLW). (A) TPP fabrication process and optical images of biodegradable soft helical microswimmers. [34] John Wiley & Sons. © 2020 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (B) TPP-based 3D printing of the microswimmers. Reproduced from [162]. CC BY 4.0.

microscopic-scale architectures with an ultrahigh spatial resolution [160, 161]. These printed 3D templates can be endowed with predesigned actuation or navigation modes only after by means of tightly adhering superparamagnetic nanoparticles (e.g. Fe_3O_4, Ni) via the above-mentioned GLAD or TAED. Based on the TPP process, Wang *et al* [31] investigated the propulsion behavior of helical microrobots with different surface coatings ranging from hydrophilic to hydrophobic. The photoresist IPL-780 was laser-written on a glass substrate to fabricate helical bodies and then the helical structures were covered with Ni for magnetic actuation and Au for antioxidation and better adhesion to following surface modification sequentially via physical evaporation. In a similar manner, Dong *et al* [34] used TPP-based 3D laser lithography and water dispersion to fabricate helix-shaped microstructures with biodegradable soft hydrogel. Whereafter, the printed helical microstructures were coated with magnetoelectric nanoparticles serving as motile components activated by a rotating magnetic field (figure 10(A)). Despite high propulsion efficiencies under a rotating magnetic field [46], the cargo transport capabilities of helical microrobots are still restricted due to their small surface areas. To this end, Ceylan *et al* [162] optimized the common architecture from single- to double-helical shape. They used the TPP process to remodel the superparamagnetic iron oxide-based matrix, which highly increased the volume-to-surface ratio (figure 10(B)). Furthermore, porous-related structures providing more net spatial space have been emergently investigated. The novel burr-like porous spherical structures designed by Sun and co-workers [44] have demonstrated higher cell delivery efficiency *in vitro* and *in vivo*. The porous spherical structures are firstly fabricated by TPP process, and subsequently depositing Ni for achieving magnetic actuation and Ti for ensuring biocompatibility by the commercially available sputtering device. Kim *et al* [68] fabricated a magnetic microrobot with many microgrooves for cell adsorption that transports cultured neural cells to bridge the disconnected neural clusters. The microgrooves with the size of axons and dendrites were engraved by TPP-based DLW with high fabrication resolution (2-μm depth and 5-μm width), in which microgrooves were designated to align the neurites for facilitating the following linkage.

4. Applications of magnetically actuated micro/nanorobots

4.1. Targeted drug delivery

Drug therapy is the most widely used treatment in clinical medicine. There are two traditional drug delivery methods: one is the intestinal drug delivery by oral administration, and the other is the blood drug delivery by injection [163]. However, both two methods belong to passive drug transport *in vivo*, which maybe lead to excessive dose and off-targeted delivery resulting in side effects of human bodies [164, 165]. The emergence of external field-driven medical micro/nanorobots has offered an excellent platform for accurately and efficiently delivering therapeutic drugs to the diseased parts of body [101]. It is worth noting that magnetically actuated micro/nanorobots are one of the most promising targeted drug delivery systems due to their remote, precise

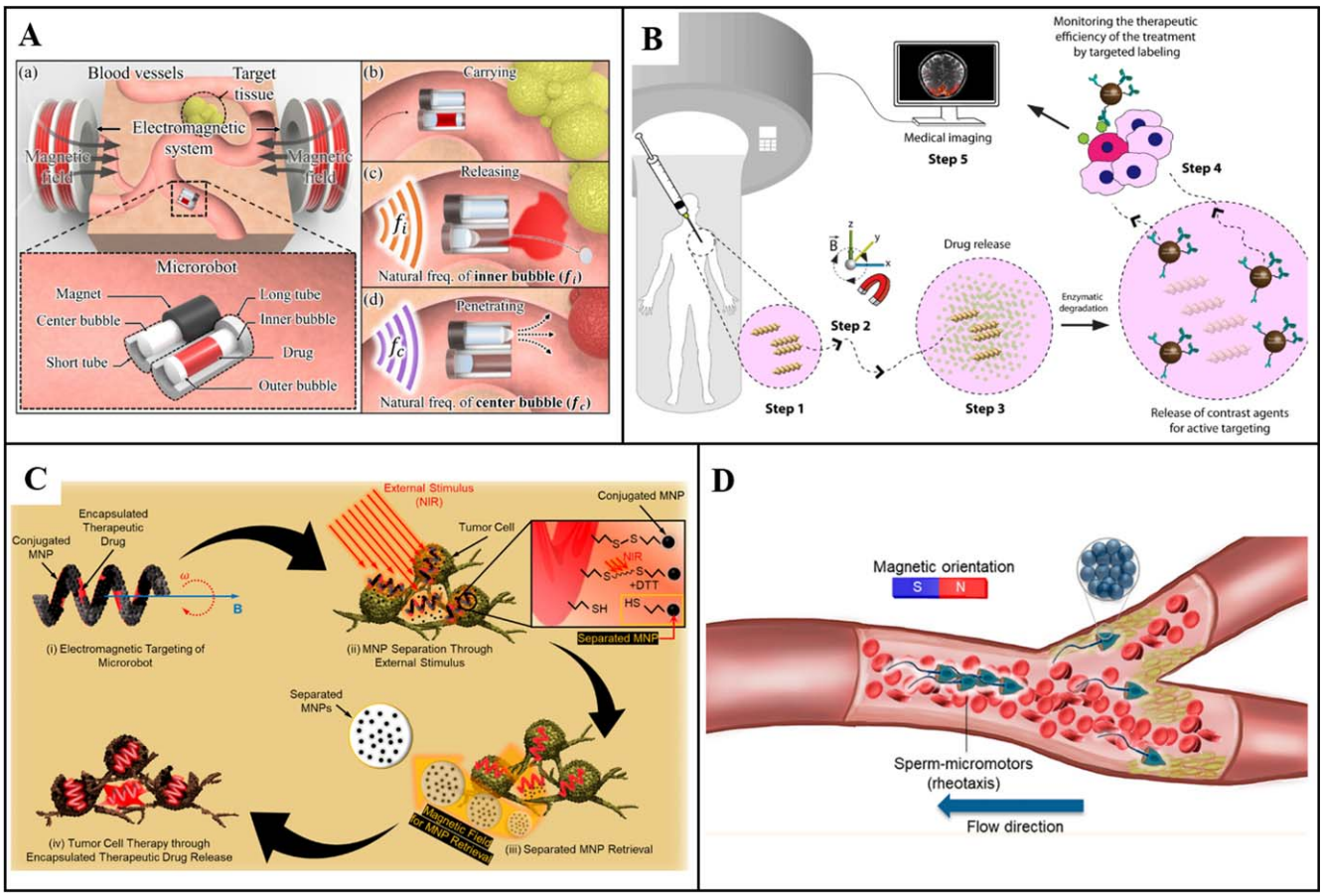

Figure 11. Targeted drug delivery. (A) Electromagnetically actuated microrobot for acoustic-triggered targeted drug delivery. Reprinted from [172], © 2020 Elsevier B.V. All rights reserved. (B) Double-helically biodegradable microswimmers for enzyme-controlled theragnostic application. Reproduced from [162]. CC BY 4.0. (C) Conceptual diagram of helical microrobots with MNP retrieval. Reprinted with permission from [58]. Copyright © 2021, American Chemical Society. (D) Sperm-micromotors in flowing blood under the magnetic navigation. Reprinted with permission from [173]. Copyright © 2020, American Chemical Society.

and non-invasive maneuverability [166, 167]. The most important step to achieve targeted drug delivery for magnetic micro/nanorobots is the drug loading or capturing. There are two main ways to load or capture drugs: one is to encapsulate the drugs inside the robot bodies, and the other is to attach the drugs on the surfaces of robot bodies [5]. So far, a large number of advanced artificial magnetic nanomaterials and biogenic materials have been developed for encapsulating or adsorbing the therapeutic drugs [162, 168–171]. Here, recent progresses in magnetically actuated micro/nanorobots for targeted drug delivery are introduced.

Two microtubes with different lengths were designed as the microrobot systems for targeted drug delivery, in which the shorter microtubule only filled the gaseous bubbles and a micro permanent magnet was installed at the end for magnetic actuation, while the longer microtubules filled drugs and the end was sealed with bubbles. Under an electromagnetic field, the microtube robot moved to the targeted position, then the bubbles at the outer end of longer microtube were removed by the acoustic excitation and the drugs could be released at the targeted position (figure 11(A)) [172]. The biodegradable microswimmers developed by Ceylan *et al* were capable of quickly responding to the concentration of MMP-2 enzyme, which boosted the release of drugs embedded in the microswimmers by swelling the hydrogel-based helical body (figure 11(B)) [162]. Anticancer drugs (doxorubicin, DOX) and magnetic nanoparticles (MNPs) were attached on the surfaces of helical microrobots for delivering DOX to the regions of cancer cells under a rotating magnetic field. After completing DOX delivery, MNPs on the surfaces of robots were separated by using dithiothreitol (DTT), and then MNPs were recovered with another magnetic field (figure 11(C)) [58]. Chen *et al* [59] proposed the spherical-shaped child-parent microrobots (CPMs) loaded DOX, which could entered into the small intestine by oral administration, and moved to the targeted area to release DOX. Active bacteria drove RBC microswimmers loaded drug molecules to complete targeted DOX transportation based on the magnetic navigation [153]. Moreover, the micromotors actuated by sperms carried cargos to move through the flowing blood under the magnetic navigation, as shown in figure 11(D) [173].

4.2. Minimally invasive surgery

In the light of motile miniaturized structures of magnetically powered micro/nanorobots, they are commonly regarded as promising candidates for various biomedical applications,

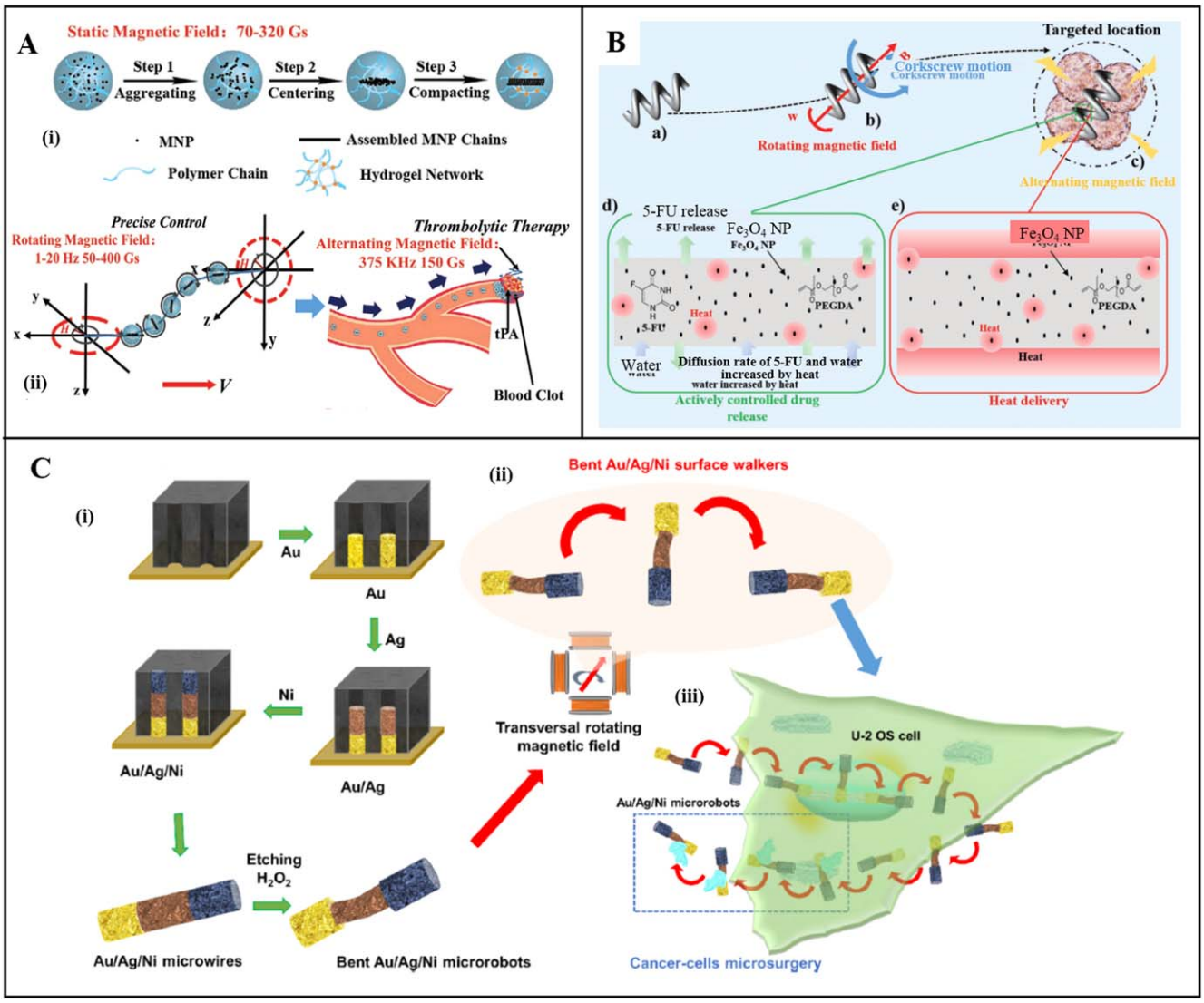

Figure 12. Minimally invasive surgery. (A) The biomimetic magnetic microrobots with magneto-collective control for targeted thrombolytic therapy. [177] John Wiley & Sons. © 2020 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (B) Propulsion mechanism and functionalities of biodegradable hyperthermia microrobots. [69] John Wiley & Sons. © 2019 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (C) Application of Ni/Ag/Au microrobots for cancer cell microsurgery. Reprinted with permission from [35]. Copyright © 2020, American Chemical Society.

especially in minimally invasive surgery [174, 175]. External magnetic actuation mode possesses natural superiorities for providing forward propulsive force or torque through thick biological tissues. Moreover, certain micro architectures may facilitate the execution of precise navigation or cell invasion in complex and constrained *in vivo* environments. To this end, Kim *et al* [176] 4D printed the soft threadiness robots with the composites composed of silicon rubber and hard magnetic microparticles. Under an applied magnetic field, the soft robots could navigate in realistic *in vitro* and *in vivo* environments such as peripheral artery and its aneurysms with localized dilation. The introduced bending stiffness variation of soft robots enabled getting through vasculature-like tortuous paths upon active steering. Meanwhile, hydrogel-based skin for reducing the contact friction could make it more efficient to generate the translational displacements. In a similar work, Xie *et al* [177] took inspiration from magnetotactic bacteria (MTB) and developed the magnetic microrobots based on their magnetosome structures. The aligned iron oxide nanoparticle (MNP) chains inside biocompatible microgel shells could serve as the navigation and propulsion components by magneto-collective control, while the swarm fashion further enhanced the efficiencies of minimally invasive targeted thrombolysis (figure 12(A)).

In order to maximize the therapeutic effect of non-invasive surgery, it should be considered to release different concentrations of drugs at the targeted positions. For this reason, the helical hyperthermia microrobots recently reported by Park *et al* [69], as shown in figure 12(B), were highlighted with actively controlled drug release capability in different modes. Furthermore, the drilling feature of helical-shaped microrobots could be set for penetrating the biopolymeric network of vitreous humor and approaching the retina for over centimeter distances [64], which moved further toward

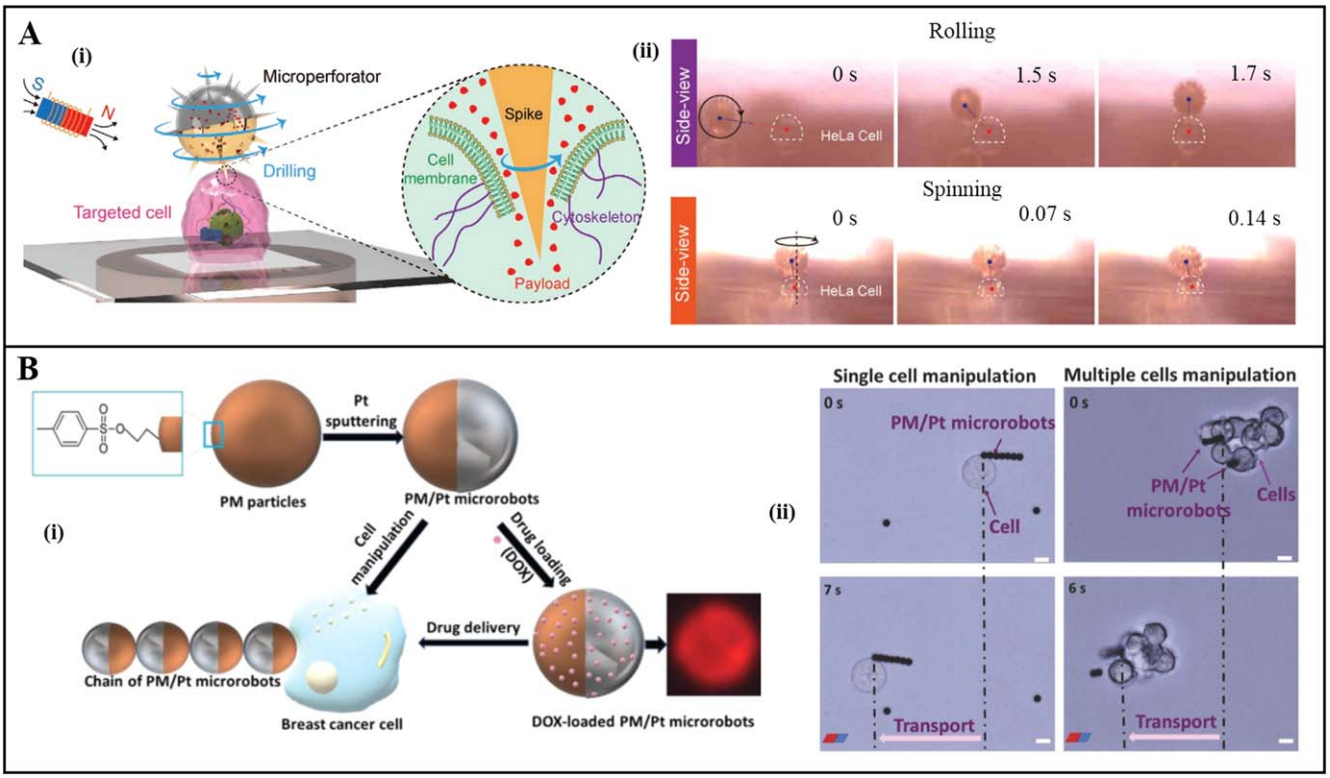

Figure 13. Cell manipulation. (A) Magnetically driven microperforator for direct cell delivery by cell membrane perforation. [179] John Wiley & Sons. © 2020 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (B) Preparation and breast cancer cells delivery of PM/Pt microrobots. [71] John Wiley & Sons. © 2018 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

targeted positions in noninvasive ophthalmological therapies. Au/Ag/Ni wire-like microscalpels [35] functionalized with a high rigidness were also capable of serving as microdrillers with slightly bent to penetrate into cancer cells. And they performed a series of operations like a section of the cytosol inside capture with cytoplasmic membrane intact while exiting the abnormal cells (figure 12(C)).

4.3. Cell manipulation

The application of magnetically actuated micro/nanorobots in cell manipulation refers to the precise mechanical manipulations of cells such as separating, capturing, transporting and patterning without changing the normal physiological properties of cells [5]. Up to now, many magnetic micro/nanorobots have been developed for cell manipulation. Here, the recent progresses of magnetic micro/nanorobots for cell manipulation are presented. The targeted delivery efficiency of Mesenchymal stem cells (MSCs) for knee cartilage regeneration can be enhanced by using magnetic micro/nanorobots [178]. The magnetic microrobots were fabricated by PLGA microfolding and surface adsorption of microclusters. MSCs were firstly loaded inside the magnetic microrobots, and then transported to the cartilage defect under the control of an electromagnetic system. MSCs would differentiate into new cartilage tissue to achieve the purpose of treatment.

In order to complete the zygote intrafallopian transfer (ZIFT), Lukas *et al* [70] developed a rotating spiral microrobot to capture, transport and release zygotes. This microrobot included two spiral structures, one was based on a helical corkscrew propeller, and the other was a spiral-shaped micropropeller. When the micropropeller rotated forward, it could enclose to protect cargos, and when the spiral pill reversed, it could release cargos. The microperforator could rotate and drill holes under the intervention of an external magnetic field to destroy the cancer cell membrane for accurately and adequately inputting drugs into the target cells (figure 13(A)) [179]. Xu *et al* reported the transport procedures of multiple motile sperm cells by magnetic micromotors. The local vortex generated by the rotation and translation movement of microcarriers under the rotating magnetic field was used to capture sperm cells. The microcarriers were enzymatically hydrolyzed by the protease to release sperm cells after transporting them to the destination [180]. Villa *et al* [71] synthesized a kind of superparamagnetic microspheres whose half of the surface was sputtered with Pt for catalyzing the decomposition of hydrogen peroxide to provide power for microrobots. The reported PM/Pt microrobots could bind amino groups directly delivery to breast cancer cells due to the reactive nature of p-toluene sulfonyl groups on their surface (figure 13(B)). Moreover, Kim *et al* [68] developed the neuron-loaded microrobots that actively connected nerve

clusters by precisely transporting nerve cells under an external magnetic field.

5. Summary, current challenges and future works

5.1. Summary

Over the past decade, researchers have made considerable achievements in the design and fabrication of multifarious magnetically actuated micro/nanorobots for biomedical applications. These micro/nanorobots can complete the important tasks in some complex biomedical environments in a non-invasive way. This article has systematically reviewed the recent breakthrough advances of magnetically actuated micro/nanorobots from the following three aspects: design strategies, fabrication techniques and applications. Main design strategies of magnetic micro/nanorobots have been discussed, including helical robots, surface walkers, ciliary robots, scaffold robots and biohybrid robots. For each design strategy, we have introduced some representative robots. Common fabrication methods of magnetically actuated micro/nanorobots have also been discussed, and followed by the latest application trials of these robots in the field of targeted drug delivery, minimally invasive surgery and cell manipulation.

5.2. Current challenges

Although a variety of impressive magnetically actuated micro/nanorobots have been successfully developed, most of the reported robots so far are only at the proof-of-concept stage, capable of simple functional demonstrations in artificially designed simulation environments. There is still a long way to go before it can be practically applied in complex biomedical environments.

Firstly, artificial micro/nanomaterials and bio-materials used to manufacture magnetically actuated micro/nanorobots need to strictly meet the requirements of biocompatibility, biodegradability, low- or non-toxicity, processability and high economic efficiency. The types and properties of raw materials required for the synthesis of existing materials are far from meeting the above requirements, and their chemical synthesis processes are extremely complex and expensive, which inevitably limits the development race of micro/nanorobots.

Secondly, the optimized 3D morphologies and structure designs are essential for enhancing the movement efficiency of magnetically actuated micro/nanorobots. However, the current micro/nano manufacturing techniques and assembly processes hardly produce the micro/nanorobots that meet corresponding design needs with high resolution, low cost and harmlessness to the environment. The existing fabrication techniques more or less have their own defects. For example, GLAD [32, 64, 65, 139–141] based physical vapor deposition technology can produce complex shape micro/nanorobots, but its preparation environment is extremely strict, and maybe produce harmful products to the environment. Although TAED [66, 89, 147–151] and magnetic self-assembly technology [131–134] are simple and environmentally friendly, the fabricated shapes and structures of micro/nanorobots are only a few. 3D direct laser writing technology [34, 44, 68, 162] can manufacture micro/nanorobots of any shape and structure, but the printing equipment is expensive and the materials available for 3D printing are very limited. Biohybrid technology [47, 49, 54, 67] is limited by biomaterials because of quite few available biomaterials.

Thirdly, due to the complex biomedical environment, the use of one or several micro/nano robots hardly meets the needs of practical applications. In order to enhance the work efficiency of magnetically actuated micro/nanorobots, swarms or clusters of micro/nanorobots can effectively complete complex tasks through their team cooperation. However, precise control of behaviors of swarms or clusters is still a challenge for researchers in the future.

Fourthly, the precise maneuvering and real-time monitoring of magnetic micro/nanorobots in real *in vivo* environments are crucial for the safe completion of biomedical tasks. To meet this demand, researchers have to develop multi-degree-of-freedom, high-precision magnetically driven systems and advanced clinical imaging systems, as well as magnetic field theoretical modeling methods and advanced control theories.

5.3. Future work

To realize the practical application of magnetic micro/nanorobots in complex biomedical environments in the future. Researchers need to make breakthroughs in the following aspects: (i) new micro/nano magnetic materials, printable materials and biomaterials need to be further explored to improve the biocompatibility and minimize the toxicity of the artificial magnetic micro/nanorobots in biomedical environments; (ii) developing new multi-material additive manufacturing strategies to realize the integrated manufacturing of magnetic nanomaterials and robot bodies. For example, the collaborative manufacturing strategy of electrohydrodynamic (EHD) printing and direct laser writing (DLW) technology may be expected to break through the bottleneck of existing micro/nano manufacturing technologies; (iii) the combination of multiple actuation modes maybe achieve more efficient and accurate control of magnetically actuated micro/nanorobots in the biological body; (iv) by combining with the clinical imaging technology, magnetic field theoretical modeling and intelligent control theory, to realize closed-loop feedback control *in vivo* environment.

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Data availability statement

No new data were created or analysed in this study.

Declaration of competing interest

All authors declare that they have no competing financial interests.

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